

Every Nontrivial Finite Product-Tree Inclusion Is Strict

Candidate solution packet for arXiv:1907.11476

9 August 2026

Status. Candidate full solution, likely valid pending expert review. The packet completely classifies the inclusions in equation (7) of the source paper.

1 Source question

Vergara studies positive definite radial kernels on homogeneous trees and their finite Cartesian products [1]. Here $\mathcal{T}_q$ is the $(q + 1)$ -regular tree, and $\mathcal{R}_+(X)$ is the cone of functions $\varphi : \mathbb{N} \rightarrow \mathbb{C}$ for which $(x, y) \mapsto \varphi(d_X(x, y))$ is positive definite on X .

For $q' \geq q$ and $N' \geq N$, isometric embeddings give

$$\mathcal{R}_+(\mathcal{T}_{q'}^{N'}) \subset \mathcal{R}_+(\mathcal{T}_q^N). \quad (1)$$

Immediately after equation (7), on page 6, the paper asks whether some of these inclusions are equalities and says that a complete answer is not known. The source passage is reproduced below.

6

I. Vergara

Again, since there is an isometric inclusion $\mathcal{T}_q^N \hookrightarrow \mathcal{T}_{q'}^{N'}$ for $q' \geq q$ and $N' \geq N$, we know that

$$\mathcal{R}_+(\mathcal{T}_{q'}^{N'}) \subset \mathcal{R}_+(\mathcal{T}_q^N). \quad (7)$$

It seems natural to ask whether some of these inclusions are actually equalities. We do not have a complete answer to this question, but Theorem 1.4 allows us to conclude the following.

Corollary 1.7. *For all $2 \leq q < \infty$, we have*

$$\mathcal{R}_+(\mathcal{T}_q \times \mathcal{T}_q) \subsetneq \mathcal{R}_+(\mathcal{T}_q).$$

We make one final remark concerning this question. The equivalence (a) $\Leftrightarrow$ (c) in Theorem 1.4 associates, to each element of $\mathcal{R}_+(X)$, a Borel measure on $[-1, 1]^N$, but not conversely. Indeed, the left-hand side of (6) depends only on the sum $n_1 + \dots + n_N$, and this imposes severe restrictions on the measure μ . To illustrate this fact, suppose that $\varphi : \mathbb{N} \rightarrow \mathbb{C}$ satisfies

$$\varphi(n_1 + n_2) = \int_{[-1, 1]^2} t_1^{n_1} t_2^{n_2} d\mu(t_1, t_2), \quad \forall (n_1, n_2) \in \mathbb{N}^2,$$

for some positive measure μ on $[-1, 1]^2$. Then, by computing the integral of $t_1^{n_1} t_2^{n_2} (t_1 - t_2)^2$, one can show that μ must be supported on the diagonal of $[-1, 1]^2$. This is actually another way of proving that $\mathcal{R}_+(\mathcal{T}_\infty^2) = \mathcal{R}_+(\mathcal{T}_\infty)$. This suggests that a more thorough analysis of the identity (6) could lead to a better understanding of the set $\mathcal{R}_+(\mathcal{T}_q^N)$ and the inclusions (7).

1.1. Organisation of the paper

The source proves strictness when $N = N' = 1$ and $q' = q + 1$, and when $N = 1, N' = 2, q' = q < \infty$. It also observes that $\mathcal{R}_+(\mathcal{T}_\infty^N) = \mathcal{R}_+(\mathcal{T}_\infty)$ for every N .

2 Proof intuition

There is no need to analyze the full moment representation. Use the sparsest nontrivial radial function: it is 1 at distance zero, t at distance one, and 0 thereafter. On a graph X , its kernel operator is exactly $I + tA_X$, where A_X is the adjacency operator.

For a finite-degree product of trees,

$$\|A_{\mathcal{T}_q^N}\| = 2N\sqrt{q}.$$

The graph is bipartite, so the bottom of the spectrum is the negative of this number. Hence the sparse kernel is positive definite precisely when $|t| \leq (2N\sqrt{q})^{-1}$. This threshold strictly decreases whenever either q or N increases, providing a witness between every two distinct finite parameter pairs. If the larger tree has infinite degree, any nonzero t already fails on a sufficiently large finite star.

3 The complete classification

We order $\{2, 3, \dots, \infty\}$ in the usual way.

Theorem 1 (All inclusions in (1)). *Let $q' \geq q$, $N' \geq N \geq 1$, and let q, q' belong to $\{2, 3, \dots, \infty\}$.*

1. *If $q = q' = \infty$, then*

$$\mathcal{R}_+(\mathcal{T}_{q'}^{N'}) = \mathcal{R}_+(\mathcal{T}_q^N) = \mathcal{R}_+(\mathcal{T}_\infty).$$

2. *If $(q', N') = (q, N)$, equality is tautological.*

3. *In every remaining case, necessarily $q < \infty$, and the inclusion*

$$\mathcal{R}_+(\mathcal{T}_{q'}^{N'}) \subset \mathcal{R}_+(\mathcal{T}_q^N)$$

is strict.

For finite q, q' , an explicit separating function is

$$\varphi_t(0) = 1, \quad \varphi_t(1) = t, \quad \varphi_t(n) = 0 \quad (n \geq 2), \tag{2}$$

where

$$\frac{1}{2N'\sqrt{q'}} < t \leq \frac{1}{2N\sqrt{q}}. \tag{3}$$

If $q' = \infty$, any $0 < t \leq (2N\sqrt{q})^{-1}$ separates the cones.

We prove the adjacency estimate first, including the endpoint.

Lemma 2 (Adjacency norm). *For finite $q \geq 2$ and $N \geq 1$, the adjacency operator of the Cartesian product satisfies*

$$\|A_{\mathcal{T}_q^N}\| = 2N\sqrt{q}. \tag{4}$$

Its spectrum is symmetric about zero.

Proof. First consider one tree and fix a root o . Put

$$h(x) = \left(1 + \frac{q-1}{q+1}d(o, x)\right) q^{-d(o, x)/2}.$$

A direct check at the root and at every other vertex gives $A_{\mathcal{T}_q}h = 2\sqrt{q}h$. The weighted Schur estimate

$$2|u(x)u(y)| \leq \frac{h(y)}{h(x)}|u(x)|^2 + \frac{h(x)}{h(y)}|u(y)|^2$$

summed over edges yields $|\langle Au, u \rangle| \leq 2\sqrt{q} \|u\|_2^2$ for every finitely supported u . Thus $\|A_{\mathcal{T}_q}\| \leq 2\sqrt{q}$.

For the reverse bound, let $u_R(x) = q^{-d(o, x)/2}$ for $d(o, x) \leq R$ and $u_R(x) = 0$ otherwise. Since the sphere of radius $n \geq 1$ has $(q+1)q^{n-1}$ vertices,

$$\|u_R\|_2^2 = 1 + R \frac{q+1}{q}, \quad \langle Au_R, u_R \rangle = 2R \frac{q+1}{\sqrt{q}}.$$

Their quotient tends to $2\sqrt{q}$.

On the N -fold product,

$$A_{\mathcal{T}_q^N} = \sum_{j=1}^N I \otimes \cdots \otimes A_{\mathcal{T}_q} \otimes \cdots \otimes I.$$

The triangle inequality gives the upper bound $2N\sqrt{q}$. The tensor vectors $u_R^{\otimes N}$ have Rayleigh quotients equal to N times the one-tree quotient, proving equality in (4). Finally, multiplication by $(-1)^{\sum_j d(o, x_j)}$ conjugates the adjacency operator to its negative, so the spectrum is symmetric. $\square$

Lemma 3 (Exact sparse-kernel threshold). *For finite q and $N \geq 1$, the function in (2) belongs to $\mathcal{R}_+(\mathcal{T}_q^N)$ if and only if*

$$|t| \leq \frac{1}{2N\sqrt{q}}. \quad (5)$$

For every $N \geq 1$ and every nonzero t , it does not belong to $\mathcal{R}_+(\mathcal{T}_\infty^N)$.

Proof. For any graph X , the kernel associated with (2) is entrywise

$$K_t = I + tA_X. \quad (6)$$

For a finite-degree product, positive definiteness of the kernel is equivalent to nonnegativity of (6) on finitely supported vectors. Lemma 2 and spectral symmetry show that this holds exactly when $1 - |t|2N\sqrt{q} \geq 0$, proving (5). Equivalently, failure above the threshold can be seen directly by applying the parity sign to the tensor approximate extremizers in the proof of Lemma 2.

For $\mathcal{T}_\infty^N$, fix all but one coordinate and select a vertex together with M distinct neighbors in the remaining coordinate. The corresponding principal matrix is $I + tA_{K_{1,M}}$, whose eigenvalues include $1 - |t|\sqrt{M}$. This is negative for sufficiently large M whenever $t \neq 0$. $\square$

Proof of Theorem 1. The infinite-degree equality is Proposition 1.5 and the paragraph following it in the source [1]; identical parameters give the second case.

Suppose now that $q < \infty$ and the parameter pairs are distinct. If q' is finite, monotonicity and at least one strict parameter inequality give

$$2N'\sqrt{q'} > 2N\sqrt{q}.$$

Choose t as in (3). Lemma 3 puts φ_t in the cone on the right of (1) but not in the cone on the left, so the inclusion is strict. If $q' = \infty$, choose any positive t satisfying the finite threshold for (q, N) ; the second part of Lemma 3 excludes it from the larger product. These cases exhaust (1). $\square$

4 A slightly more general separation

The same proof gives

$$\left\| A_{\tau_{q_1} \times \dots \times \tau_{q_N}} \right\| = 2 \sum_{j=1}^N \sqrt{q_j} \quad (7)$$

when all q_j are finite. Thus the sparse witness separates any coordinatewise proper inclusion between anisotropic finite products as well. This is not needed for Theorem 1, but shows that the mechanism is intrinsic to products rather than to equal factor degrees.

5 Verification, limitations, and novelty

The proof uses only finite-support quadratic forms. In particular, it does not depend on the product moment representation or on convergence of infinite matrices. The endpoint in (5) is included by the explicit upper bound, while failure beyond it is witnessed by finitely supported approximate extremizers. The optional script `code/verify_sparse_witness.py` checks the displayed Rayleigh formulas and sample separating parameters; it is not part of the proof.

A bounded literature search on 9 August 2026 used the run’s cheap indexes, the exact title and arXiv id, the author name, the displayed inclusion, and the phrases “positive definite radial kernels,” “products of homogeneous trees,” “strict inclusion,” and “adjacency operator.” It found the source and related radial Schur-multiplier background, but no later result giving this classification or the witness (2). Novelty confidence is moderate pending specialist review.

6 Human-review recommendation

The packet is ready for expert review as a candidate full answer to equation (7). The decisive checks are the weighted Schur estimate in Lemma 2, the tensor-product lower bound, and the equivalence (6) at the endpoint. The literature status should also be checked by a specialist in radial multipliers on trees.

References

- [1] I. Vergara, *Positive definite radial kernels on homogeneous trees and products*, arXiv:1907.11476v2 (2019); J. Operator Theory **84** (2020), 435–460.